e61 fiscal model - updated research prototype

Updated 26 October 2026. This version implements the first four development priorities from the original prototype and adds a dedicated Revenue tab. The previous code, inputs, README and PDF are preserved in `archive/v1`.

Open http://127.0.0.1:7461 while the local server is running. To restart from the parent `projections` directory:

```
& 'C:\Program Files\R\R-4.5.1\bin\Rscript.exe' --vanilla 'e61 fiscal model\run_app.R'
```

The app uses shiny, bslib and ggplot2. The projection engine itself uses base R. It runs offline from cached sources. Scenario changes begin in FY2030-31, after the common FY2029-30 PBO anchor. Years label financial years by ending year. All monetary model amounts are AUD billions unless otherwise stated.

This remains a research scenario model, not a validated e61 forecast or a full household microsimulation. New evidence replaces several arbitrary inputs, but programme cost responses and representative pension wealth remain important assumptions. Steps 5-6, historical validation and external review, remain open.

What changed in steps 1-4

Step	Implemented	Remaining qualification
1. Source vintages and coverage	2026-27 PBO workbook; 2026 IGR population, participation and cost profiles; ABS tax and GFS reconciliation; Commonwealth/state/national bridge	No separate long-run budgets for each jurisdiction; local coverage and transfer elimination appear together in the net bridge
2. Labour and service exposures	Observed ABS age/sex participation, unemployment and hours; IGR participation and average-hours projections; sourced health and aged-care age-cost profiles	Annual age distributions are interpolated; participation uses an aggregate-target bridge, not Treasury's full cohort model; education remains a proxy
3. Care and pensions	Separate NDIS and aged-care modules; NDIS cohorts retained after 65; pension age 67, income/assets tests, deeming and super accumulation	Pension wealth archetypes and care intensity remain stylised; not all 2026 care reforms or eligibility decisions are simulated
4. Tax shares and elasticities	Observed ABS tax heads, ATO income percentiles, actual resident tax schedule and LITO; explicit nominal/CPI/wage threshold rules; historical elasticity estimates	Short-sample elasticities are associations; taxable-income cells are not individual records; some offsets and state tax rules remain outside scope

The new Revenue tab explains the annual revenue path, composition, personal tax thresholds, bracket creep, calibration and estimated elasticities. Annual revenue detail can be downloaded separately from the full fiscal projection.

Deploying Fiscal_tester

`scripts/deploy_fiscal_tester.R` follows the calculator deployment scripts and uses the locally configured `e61-institute` account on `shinyapps.io`. It does not contain credentials. From the parent `projections` directory, check the runtime bundle without uploading:

```
& 'C:\Program Files\R\R-4.5.1\bin\Rscript.exe' --vanilla 'e61 fiscal model\scripts\deploy_fiscal_tester.R'
```

Publish or update the app:

```
& 'C:\Program Files\R\R-4.5.1\bin\Rscript.exe' --vanilla 'e61 fiscal model\scripts\deploy_fiscal_tester.R' --deploy
```

The defaults are `--account=e61-institute --server=shinyapps.io --app-name=Fiscal_tester --app-title="e61 Fiscal Tester"`; each can be overridden. The URL is https://e61-institute.shinyapps.io/Fiscal_tester/. From an R console in `e61 fiscal model`, a check can also be run with `source('scripts/deploy_fiscal_tester.R')`.

The explicit file list includes model code, the required CSV extracts and the source URL manifest used by the assumptions download. Raw workbooks, reports, archives, test code, logs and local output files are excluded. The script stages and starts that exact file set to catch missing inputs before uploading, checks UTF-8 source and finite default projections, and records the file/package lists in `outputs/deployment_files.txt` and `outputs/deployment_dependencies.csv`. Deployment records are kept in `rsconnect/shinyapps.io/e61-institute/` for future updates. Remote package resolution and deployment require internet access.

Starting data and calibration

`scripts/acquire_update_sources.R` caches primary sources and records URLs, checksums and the snapshot date in `data/raw/v2/download_manifest.csv`. `scripts/prepare_update_inputs.R` extracts the model inputs offline into `data/processed`. Acquisition retains existing cache files; archive a source before intentionally refreshing its vintage. No parent project files are edited.

Principal source vintages:

- PBO 2026-27 National Fiscal Outlook, August 2026: consolidated national, Commonwealth and state fiscal series to 2030.
- Treasury 2026 IGR, September 2026: public chart workbooks, including population, structural participation, average-hours assumptions and age-cost profiles. IGR fiscal benchmarks cover the Commonwealth, not all Australian governments.
- Centre for Population 2025 Population Statement age/sex workbook, published January 2026: sex ratios within each age, used alongside IGR population totals and age shares.
- ABS Government Finance Statistics 2024-25 and Taxation Revenue 2024-25, released in 2026: observed consolidated expenses, taxes and non-tax revenue. The GFS workbook already existed in the project; it is copied into this snapshot.
- ABS Labour Force Detailed March 2026 Table 1 and EM1a: a matched 12-month April 2025-March 2026 window for participation, unemployment and hours by age and sex. This is a matched detailed-data vintage, not a claim that March is the latest headline labour release.
- ATO Taxation Statistics 2023-24, Individuals Table 16A, downloaded from the ATO dataset on data.gov.au: actual taxable-income percentile counts and income sums.
- NDIA Annual Report 2024-25 Note 1.1G: $46.352178bn participant plan expenses, excluding administration; Q4 2024-25 Table D.13: participation rates by age.
- AIHW GEN spending on aged care, August 2026: $39.8bn all-government expenditure in 2024-25.
- ANAO Age Pension audit: $62.2bn 2024-25 outlay, used as a cash-to-GFS allocation approximation.

- Services Australia, DSS and the legislated September 2026 deeming determination: pension rates and means-test settings, recorded with URLs and effective dates in `data/policy_parameters.csv`.

Fiscal coverage and reconciliation

PBO national totals already consolidate Commonwealth, state/territory and local general government. The model uses these totals directly. It does not sum gross Commonwealth and state budgets or count GST grants and NDIS funding contributions as extra national revenue or expenses.

`consolidation_bridge.csv` reports Commonwealth, state and national revenue and expenses. Its residual combines transfer elimination, local-government coverage and accounting adjustments; **it is not a separately identified transfer total**. The app displays this bridge. Independent long-run jurisdictional forecasts have not been constructed.

Detailed tax tables have a small revision/consolidation difference from national GFS. Their tax heads are reconciled proportionately to observed consolidated GFS taxation revenue. `tax_consolidation_bridge.csv` preserves both reported intergovernmental taxes and the actual net adjustment. GST is collected once in the Commonwealth tax data, with no state GST grant added.

PBO dollar amounts control the 2030 fiscal anchor. Nominal GDP is inferred from published gross debt dollars and its GDP ratio. Rounded reported ratios can therefore differ slightly from dollar-derived spending and revenue shares. The model preserves the dollar accounting identities rather than forcing all rounded ratios to agree simultaneously.

Spending allocation

2025 GFS primary expenses are allocated to eight modules: Health, Education, Age Pension, Aged Care, NDIS, Other Social Protection, Other Public Services and Defence. Conventional interest is removed once. The pension, aged-care and NDIS amounts are **subtracted from Social Protection**, not added above the GFS total. The broad functional mapping remains an approximation across programme and GFS reporting concepts; the reconciliation does not establish exact classification equivalence. NDIA administration stays in the residual public-service envelope.

Source exposures and default cost assumptions carry these allocations to 2030; one common factor reconciles them to PBO primary expenses. These are model allocations, not published PBO programme forecasts. Scenario changes never recalibrate this starting point. `outputs/programme_calibration.csv` exposes the 2025 values, raw calibration factor and 2030 amounts.

Demographics, participation and hours

IGR C8.6 publishes single-year age shares at 2025, 2036 and 2066. The model linearly interpolates those shares annually and normalises the published rounded shares. It chains annual IGR C8.1 population growth from the 2025 IGR population level. Centre for Population male/female ratios allocate each age to sex. The result is a reconciled age/sex input, **not an independently estimated population cohort-component forecast**. Rounded annual growth can produce a small difference from the IGR's rounded population-level chart.

ABS observed five-year age/sex participation and unemployment rates replace the old employment propensities. Annual participation probabilities shift in common log odds within each sex to reproduce IGR C9.13 structural participation targets. This is a transparent projection bridge; it does not reproduce Treasury's cohort participation model or independently identify future retirement behaviour.

Employment equals population times participation times one minus unemployment. Observed unemployment rates are held by age and sex. EM1a supplies actual hours per employed person, including non-working respondents. Detailed source rates and hours are available to 65+ as a combined group, so the model does not invent 75+ employment weights. Average hours per employed person follow the IGR C7.2 aggregate trend of -0.1% per year; the age/sex hour mix is rescaled to that target, avoiding a second composition effect. Participation and hours are common across the four main economic scenarios in this version.

```
total hours = employed people * average weekly hours * 52
real output index = total hours index * productivity index
productivity index = (1 + selected productivity growth)^years
```

Nominal output and economic controls

With linked NGDP, domestic prices grow at 2.5% annually:

```
NGDP = starting NGDP * real output index * domestic price index * ToT index^0.20
```

The terms-of-trade setting is a permanent level shift phased over ten years to 2040. The 0.20 nominal-income elasticity is a scenario assumption, not an estimated trade model. It changes nominal income without changing real production.

With an NGDP override, nominal GDP instead follows the selected growth rate and domestic prices are the residual implied by the same identity. Higher productivity at fixed NGDP means lower inflation. This is a different experiment from higher productivity with fixed inflation. Implied inflation is displayed, including deflation under extreme combinations. Negative productivity is permitted.

The domestic price index remains a common CPI/wage-deflator proxy. The GDP deflator additionally includes the ToT income effect. Hourly wages track domestic prices and productivity; average earnings also reflect average hours per worker.

Programme spending mechanisms

Audit of spending shares and the three drivers (26 October 2026)

The number, unit-cost and intensity framework is applied annually. The previous Spending chart only compared the scenario and reference in 2066, concealing changes already present in both paths. The Spending tab now adds annual paths and a driver table showing starting and ending GDP shares and the multiplicative factors behind them. No spending equations or projection parameters were changed in this audit.

At current defaults, the shares are:

Programme	2029-30 (% GDP)	2065-66 (% GDP)
Health	7.35	10.40
Aged care	1.58	3.11
NDIS	1.61	2.08
Education	4.76	4.14

Programme	2029-30 (% GDP)	2065-66 (% GDP)
Age Pension	2.24	1.52
Other social protection	5.43	5.18
Other public services	8.87	9.02
Defence	1.59	2.50

For service programmes the exact identity is:

```
spending/GDP relative to 2030 =
  number index * real unit-cost index * intensity index
  / (nominal GDP index / domestic price index)
```

Number is age-weighted demand for health and aged care, participants for NDIS, an education-age proxy for education, or the relevant population for other services. Health's age-weighted demand rises 1.56 times, compared with total population rising 1.34 times. Its real unit-cost factor is about 1.50 and intensity about 1.20, against the denominator rising about 1.98 times. The resulting share increases about 41%. Ageing and costs therefore already raise spending relative to GDP in the baseline, before changing a sidebar control.

Age Pension instead decomposes into eligible population, recipient coverage and mean payment per recipient. Mean payment also reflects means-test composition, not just statutory benefit indexation. At the default 4% real super return, reduced coverage outweighs demographic pressure. Defence is a separate GDP-share target, and is explicitly excluded from the three-driver identity.

There are two reasons the 2066 scenario/reference comparison can move little:

- Demography is the same in both paths; there is currently no scenario control for population ageing, migration, disability incidence or healthy ageing.
- Changing NGDP growth alone, with real output unchanged, changes implied inflation. All these spending modules immediately pass through domestic price growth. That same price factor appears in GDP and cancels from their GDP shares. This is an explicit model restriction, not evidence that real costs or ageing are irrelevant. Nominal spending still changes, and debt/interest can respond. Spending indexation lags and different programme price indices are not currently simulated.

Changing productivity, extra service prices or NDIS intensity can change service GDP shares. Baseline intensity trends and wage pass-through differ by programme, but are assumptions rather than estimated sector productivity relationships. The next substantive work is to validate these parameters and permit separate demographic and programme-cost scenarios, rather than force shares to respond to a purely nominal change.

`spending_driver_audit()` exposes the annual factors and reconstructs each non-target programme's nominal spending and GDP share. Validation writes `outputs/spending_driver_audit.csv` and checks both identities, faster aged health/care exposure growth, the service-price response and cancellation of a purely nominal GDP change.

```
service spending = calibrated base * demographic exposure * unit cost * intensity
unit cost = domestic prices * productivity^wage_pass_through * extra service prices
```

Health uses the sum of major-component age-cost ratios in IGR C12.10. These are Commonwealth health profiles applied as a proxy to consolidated GFS Health; they do not capture every state-funded service. Aged care uses the separate age-cost profile in IGR C6.6, whose underlying fiscal incidence data are 2023-24. Age-cost profiles are held fixed; healthy ageing and changing morbidity are not estimated.

Education uses IGR C6.6's combined Education and Other profile after subtracting its minimum per-person value as an approximation to age-neutral other spending. This is explicitly an education **proxy**, not observed enrolment by programme. Other Social Protection follows the population aged 15-66 after removing the three separately allocated programmes. Remaining services follow total population.

NDIS uses the observed age-specific participation rates from NDIA Table D.13 below 65. Starting recipients aged 65+ are retained and aged annually; there are no new entrants at 65 or older. Population cohort ratios supply demographic retention, including migration effects rather than pure survival probabilities. The top-coded 100+ group retains the previous 99 and 100+ cohorts. Costs use the calibrated programme base, with wage/price exposure and an adjustable real service intensity trend. This is not the Scheme Actuary's disability-specific entrant/exit model, and does not implement every 2026 access reform or distinguish SIL costs.

NDIS benchmark check (25 October 2026)

At default economic settings, nominal Scheme spending rises from $56.3bn in 2029-30 to $73.7bn in 2034-35, averaging 5.5% annual growth. Growth in 2034-35 itself is 5.5%. The 0.5% intensity setting is extra cost per participant above the wage/price path, not total Scheme growth. Under-65 participation rates are held fixed by age; there is no independent rollout or rising-incidence trend. The sidebar displays implied total nominal spending growth next to the extra-cost slider: compound annual growth over 2030-35 and 2035-66, plus the 2034-35 spending level. This updates with the full economic scenario. At otherwise default assumptions, 0.5% extra cost per participant implies about 5.5% annual growth in 2030-35; a 2% setting implies about 7.1%. The 2030 programme allocation is scaled with every other programme to reconcile the PBO aggregate; it is not a published programme forecast.

There are materially different official policy vintages:

- The NDIA 2024-25 actuarial report, table 1 and section 4, projects total accrual expenses of $95.755bn in 2034-35 and 7.2% growth that year. Our $73.7bn is $22.1bn (23%) lower. This older projection predates the 2026 reforms.
- The PBO July 2026 outlook, section 5.3, describes reform targets below 2% annual growth over the forward estimates and approximately 5% subsequently. These are policy-dependent targets, not an assurance the savings will be achieved.
- 2026 IGR chart C12.17 reports **total Scheme** spending of 1.5% GDP in 2029-30, 1.6% in 2034-35, 1.8% in 2049-50 and 2.0% in 2065-66. Our corresponding shares are 1.61%, 1.64%, 1.84% and 2.08%. Each uses its own GDP denominator and IGR values are rounded to one decimal. The Commonwealth net contribution in C12.5 has different coverage and must not be substituted for total Scheme costs.

Our growth is low against the older actuarial path, and broadly close to newer long-run GDP shares. That numerical proximity does not validate the mechanism: we do not explicitly simulate the reform savings, disability mix or support-cost pressures. The baseline has therefore not been retuned to force a match to an older projection. Before using it as a forecast, replace the common programme allocation with a reconciled, dated Scheme anchor and validate participant entrants/exits and cost per participant separately. Preserve a reform-delivery scenario and a higher-pressure scenario rather than assuming either is certain.

`scripts/check_ndis_and_super.R` reproduces the IGR extraction and comparisons offline from the cached workbook. It writes annual results and an explicitly dated older-vintage comparison to `outputs/ndis_benchmark_check.csv` and `outputs/ndis_2025_vintage_check.csv`. The Spending tab shows the same coverage and vintage qualifications alongside a live scenario comparison.

Wage pass-through remains a net service-cost assumption: Health .70, Education .80, Aged Care .80, NDIS .85, residual Social Protection .65 and other services .70. The extra service-price slider is above that wage-linked path. Default excess intensity growth is 0.5% for Health, 0.3% for Aged Care, 0.5% for NDIS and zero elsewhere. These parameters are **not newly estimated causal coefficients**. Defence converges to 2.5% GDP by 2035. Net capital investment grows with population, domestic prices and half of productivity growth. Depreciation is not added again.

Age Pension, means tests and retirement incomes

The fixed annual coverage-decline parameter has been removed. The replacement module uses age 67 eligibility and calculates annual entitlements from explicit income and assets tests, taking the lower payment after those tests. Singles, couples, homeowners and non-homeowners have separate published free areas and maximum rates. Deeming uses the September 2026 1.75%/3.75% rates and thresholds. The combined couple income taper is 50 cents per dollar, or 25 cents per person; the asset taper is $3 per $1,000 per fortnight for the household.

Four wealth/income archetypes represent full-rate, assets-tested, income-tested and non-recipient cases. Their initial status weights come from the **starting 2026** IGR C12.19 distribution. Future IGR coverage is retained as a benchmark, not fitted or imposed on this model. Representative assets and private incomes are constructed around the statutory thresholds; half-rate part-pension cases are assumed. Person weights of 35% single/65% couple members and 75% homeowner/ 25% non-homeowner are modelling judgements, not extracted household estimates.

For cohorts retiring after 2030, additional superannuation accumulates from 12% SG contributions, less a simplified 15% contributions tax, using the ATO mean taxable income as an earnings proxy and the selected real return (default 4%, selectable from 0% to 7% in half-percentage-point steps, including 6.5%). Balances earn returns and use a stylised 5% drawdown after retirement. This is not a reconstruction of full past work histories, contribution caps, housing wealth, residence eligibility, Work Bonus, partner-age differences or observed household wealth distributions. It improves the mechanism while leaving substantial pension calibration work before publication.

The real-return control operates **only through private retirement assets and Age Pension means tests**. Higher returns increase accumulated contributions and slow asset depletion after retirement. Larger assessable assets and deemed income reduce pension eligibility/payments; smaller deficits then reduce debt and subsequent interest costs. Deeming rates do not automatically follow the chosen return. GDP, government borrowing rates, public asset yields and the superannuation-tax revenue block do not change directly. In particular, the model does not capture extra earnings-tax receipts from higher fund returns.

With CPI-indexed tax scales and all other defaults, illustrative sensitivity is:

Real super return	Age Pension in 2065-66 (% GDP)	Net debt in 2065-66 (% GDP)
2%	2.62	111.5
3%	2.07	102.2
4% (default)	1.52	92.7

Real super return	Age Pension in 2065-66 (% GDP)	Net debt in 2065-66 (% GDP)
6.5%	0.55	74.8
7%	0.42	71.9

This large sensitivity depends on synthetic wealth archetypes, the 5% drawdown and simplified accumulation. It is a model sensitivity, not a measured fiscal effect or a validated pension forecast. Under Constant Relative Standards, means-test changes are bypassed and the return control has no fiscal effect. Annual mechanisms are checked in `tests/validate_update.R`; the reproducible sensitivity table is `outputs/super_return_sensitivity.csv`.

Maximum payments use a non-decreasing benchmark of prices and average earnings; means-test free areas follow prices. This approximates legislated indexation, without separately projecting PBLCI or the exact MTAWE benchmark. September 2026 settings are uprated to the 2030 starting point; 2030 programme expenditure is then controlled by the common fiscal calibration, not the synthetic population's unadjusted dollar total.

Under Constant Relative Standards, starting pension entitlements per eligible person are maintained relative to the payment benchmark instead of applying future wealth tests; residual Social Protection becomes fully wage-linked. This policy alternative is distinct from the indexed-rules scenario.

Revenue projections

Observed allocations and the common anchor

Fixed 42/20/25 percent revenue weights are removed. ABS observations identify personal income tax, company tax, superannuation tax, Medicare/FBT, GST, excises and customs, payroll, property, stamp duties and other taxes. GFS separately identifies royalties, fees/other non-tax revenue and interest receipts.

Observed 2025 heads are carried to 2030 under the default nominal cost path; personal income tax also reflects the change in the statutory schedule. One common reconciliation makes their sum equal PBO 2030 revenue. The calibration is frozen across scenarios. The app's detailed revenue table and downloads expose each head rather than treating all receipts as a fixed GDP share.

Personal income tax and threshold policy

ATO Table 16A supplies counts and taxable income by percentile, sex and state. The model aggregates mutually exclusive sex/state cells into 100 percentile cells and applies the resident schedule to each cell's mean taxable income. It does not sum published subtotal rows. The distribution is uprated from 2023-24 to 2030; future per-taxpayer income follows average nominal earnings, and taxpayer numbers scale with employment. This assumes a stable relative income distribution and does not independently model pensioner, non-resident or capital-gains tax bases.

The 2030 schedule uses thresholds $18,200/$45,000/$135,000/$190,000 and marginal rates 0/14/30/37/45%, including the legislated 2027-28 lower rate. LITO is explicit. Medicare and FBT are modelled in their separate observed tax block; income-tax outputs are not incorrectly labelled as including those levies. The WATO offset, whose wage-only eligibility cannot be identified from Table 16, is not simulated. Its starting aggregate effect and other differences enter the fiscal calibration.

Three annual threshold conventions are selectable from 2030:

- **Fixed nominal:** unchanged nominal thresholds and offset amounts. This corresponds to the statutory threshold convention, without assumed future cuts.
- **CPI indexed:** thresholds and LITO amounts increase with domestic prices.
- **Wage indexed:** thresholds and LITO amounts increase with average earnings.

CPI indexation is the default policy assumption, including on reset and in the fixed reference. It removes inflation-driven bracket creep while retaining bracket creep from real earnings growth. It is not current legislation; fixed nominal thresholds remain a selectable statutory-convention alternative. The tax schedule itself generates the changing effective rate; there is no 1.1 proxy income-tax elasticity. The app shows the direct revenue difference relative to wage indexation under the same economic scenario. That comparison does not re-simulate debt and interest receipts, which would require a complete financing counterfactual. No revenue/GDP cap or hidden tax relief is applied.

Other tax bases and estimated elasticities

Nine annual changes, 2017-2025, estimate each tax head's association with nominal GDP growth, using a regression with an intercept. Estimates and standard errors are retained. A Normal(1, 0.5 squared) prior shrinks noisy elasticities toward one. These are short-sample descriptive estimates, affected by policy, COVID and base measurement. They are not causal responses identified from productivity shocks.

For long-run projections the reference assumes unit-elastic growth with the relevant economic base. Estimated elasticities apply only to **scenario deviations**:

```
scenario receipts = anchor receipts * reference base * (scenario base / reference base)^elasticity
```

This avoids compounding a short-sample policy trend into baseline tax/GDP for 40 years. Neither the estimated intercept nor a regression time trend is projected. Personal income tax uses its explicit schedule instead of this regression rule.

Company, property, stamp duties and other taxes use NGDP as a base proxy. GST and excises use domestic nominal output as a consumption proxy. Payroll, Medicare/FBT and superannuation taxes follow employment and average earnings. Royalties follow domestic nominal output plus a ToT elasticity of 2, which remains an assumption. Fees and other receipts use unit-elastic domestic nominal output; dividends are inside that residual rather than modelled from separate public-corporation assets. State-specific payroll thresholds, property prices, consumption propensities, company profits, excise erosion and energy-transition effects are not separately estimated yet. These limitations remain visible in the Revenue tab.

Debt and financial assets

All spending and tax cases use the same financing engine. Interest is charged on opening gross debt. Its effective rate starts at the PBO 2030 bill divided by 2029 gross debt and converges by 2040 to the selected rate, 4.4% by default. Interest receipts use opening financial assets and the anchor-calibrated yield.

Assets maintain the initial stock per person in domestic purchasing power; acquisition is explicitly financed. After gross debt is repaid, further surpluses accumulate in interest-bearing assets. Net debt may be negative.

```
fiscal balance = revenue - expenses - net capital investment
gross debt change = -fiscal balance + financial asset acquisition
net debt = gross debt - financial assets
```

There is no automatic fiscal repair. Revaluation, maturity buckets, indexed bonds and a full public-sector balance sheet remain outside scope. Recursive identities apply from 2031; published historical/PBO stocks are not reconstructed from rounded historical flow ratios.

Interpreting the productivity and fixed-NGDP example

The Debt & interest tab compares gross debt, modelled financial assets and net debt, plus interest paid, interest received and net interest. Charts switch between dollars and GDP shares. Tables expose the selected year's opening gross debt, borrowing rate, GDP denominator, asset yield and implied real borrowing rate. The annual download includes the same diagnostics. With an NGDP override, an optional third path holds every other scenario assumption fixed and restores linked NGDP. This isolates the checkbox's effect; the fixed reference still uses 1.2% productivity and all other defaults.

With 1.7% productivity and otherwise default assumptions, in 2065-66:

Measure	Linked NGDP	NGDP fixed at 4.4%
Nominal GDP ($tn)	20.16	16.53
Gross debt ($tn)	10.22	10.00
Financial assets ($tn)	2.46	2.01
Net debt ($tn)	7.76	7.99
Net debt (% GDP)	38.49	48.34
Interest payments ($bn)	433.37	423.06
Interest payments (% GDP)	2.15	2.56

Interest payments fall in dollars, while their GDP share increases because GDP is proportionately lower. Gross debt also falls in dollars, but financial assets fall by more, so dollar net debt rises slightly. The asset path depends on the domestic price level, which changes with the NGDP override. Both scenarios still have lower debt and interest shares than the fixed default reference (92.67% net debt/GDP and 4.43% interest/GDP). Comparisons to that reference must not be confused with comparisons before and after ticking the checkbox.

The interest equation is `interest[t] = nominal rate[t] * gross debt[t-1]`. Dividing by current GDP does not make it a function of current net debt/GDP. The nominal borrowing-rate assumption is independent of inflation. Lower implied inflation therefore raises the real borrowing rate; there is no automatic Fisher adjustment, maturity structure or monetary-policy reaction. This is a modelling restriction to expose and test, not an accounting discrepancy. Likewise the model treats its financial asset stock as earning one calibrated yield; it is not a full decomposition of public-sector financial assets.

`tests/validate_debt.R` reproduces this comparison and passes 21 checks on gross and net debt, opening-stock interest, asset income, the GDP denominator and the NGDP toggle. It writes `outputs/productivity_ngdp_debt_comparison.csv` and `outputs/debt_validation_checks.csv`. `debt_diagnostics()` adds reporting fields without changing the financing or projection equations. Opening-stock fields are blank at the published 2030 anchor; recursive identities apply from 2031.

Comparisons, validation and outputs

The reference is fixed at all default assumptions: productivity 1.2%, linked NGDP, 0.3% extra service-price growth, unchanged ToT, CPI-indexed tax thresholds and default pension, care-intensity, 4% real super return and interest settings. Changing a control updates only the scenario. This corrects the earlier comparison behaviour in which selected policy settings also moved the reference and could leave the reported net-debt difference at zero.

The Revenue tab additionally isolates personal tax threshold policy within the selected economic scenario. Shapley attribution now includes the four economic groups plus a Policy and Financing group, reconciling their interactions to the total 2066 net-debt difference from the fixed reference. It is model accounting, not causal inference. Groups unchanged from defaults receive zero contribution.

Run validation from `projections`:

```
& 'C:\Program Files\R\R-4.5.1\bin\Rscript.exe' --vanilla 'e61 fiscal model\tests\validate_update.R'
```

All **88 updated engine checks** passed on 26 October 2026. Checks cover source dimensions, labour identities, care separation, fiscal calibration, observed tax totals, actual tax brackets/LITO, indexation, pension tests, controlled economic changes, debt/asset identities under stress, and attribution interactions. They establish implementation consistency, not historical predictive performance. `tests/browser_smoke.R` checks live controls, tabs, reset and downloads. `tests/validate_indexation.R` additionally verifies that wage-indexed and fixed nominal thresholds change scenario net debt while the reference and its exported series stay fixed. It checks the policy contribution and mixed-control attribution. The browser regression checks both the net-debt headline and its reference gap.

Main files:

- `R/model.R`: public entry point; `R/model_v2.R`: projection and mechanism code.
- `scripts/acquire_update_sources.R`: cache sources and archive the first version.
- `scripts/prepare_update_inputs.R`: reproducible data extraction and reconciliation.
- `data/raw/v2`: raw workbooks, reports, chart data and request manifest.
- `data/processed`: model inputs, tax elasticity estimates, coverage bridges, IGR Commonwealth benchmarks and source checksums.
- `data/policy_parameters.csv` and `data/ndis_incidence_2025.csv`: explicit manually transcribed anchors/rules with provenance. Old top-level CSVs not loaded by v2 are retained for traceability; the active inputs are named in `load_inputs()`.
- `outputs/reference_projection.csv`, `reference_components.csv`, `reference_revenue.csv`, `reference_revenue_diagnostics.csv`: default results.
- `outputs/programme_calibration.csv`, `labour_projection.csv`, `ndis_participant_projection.csv`, `pension_projection.csv`: mechanism diagnostics.
- `outputs/validation_checks.csv`, `browser_smoke_results.txt` and screenshots: verification evidence. Logs and PID identify the local Shiny process.
- `README.pdf`: printable version, regenerated by `scripts/render_readme_pdf.R`.

Next development priorities

The first four priorities now have working implementations, with the remaining qualifications above. The next priorities are:

1. Backcast programme spending, taxes and balances and assess historical forecast vintages. Use the existing top-down model as an independent challenger.
2. Replace synthetic pension wealth and family/tenure weights with household or administrative distributions; improve contribution histories, drawdowns and exact indexation before calling the pension module a current-policy forecast.
3. Add separately estimated service productivity, disability-specific NDIS costs, residential/home-care demand, education enrolment and current-policy reforms.
4. Improve tax bases, offsets, state thresholds, excise erosion and jurisdictional budgets; quantify structural and policy uncertainty before public release.